



8007 - Characterizing C60 cations in the cavity of NGC 7023

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NGC7023				
	1	MIRI-MRS 25	MIRI Medium Resolution Spectroscopy	(1) NGC7023-25
	5	MIRI-MRS-25-BKG	MIRI Medium Resolution Spectroscopy	(4) NGC-7023-Bkg-25
	2	MIRI-MRS 15	MIRI Medium Resolution Spectroscopy	(2) NGC7023-15
	6	MIRI-MRS-15-BKG	MIRI Medium Resolution Spectroscopy	(6) NGC-7023-Bkg-15
	3	MIRI-MRS 5	MIRI Medium Resolution Spectroscopy	(3) NGC7023-5
	7	MIRI-MRS-5-BKG	MIRI Medium Resolution Spectroscopy	(5) NGC-7023-Bkg-5

ABSTRACT

Buckminsterfullerene (C60) is a large carbonaceous molecule consisting of 60 C atoms, that is widespread and abundant in the universe. C60 belongs to a family of large aromatic molecules comprised of polycyclic aromatic hydrocarbons (PAHs) and fullerenes. These aromatic species play an important role in the heating and the charge balance of the interstellar medium (ISM). C60 is the only large aromatic molecule to be firmly identified in space and is thus a reference point for the studies on the behaviour of other aromatic species in space. Because of its low ionization potential and harsh conditions in the ISM, we also expect the presence of C60 cations, C60⁺ and C60⁺⁺, in the ISM that have eluded detection till date. We propose to obtain MIRI-MRS observations of 3 positions in the cavity of the reflection nebula NGC 7023, to firmly identify and characterize the infrared spectral signatures of C60⁺ and C60⁺⁺. Given its strong radiation field and well studied photodissociation region, the reflection nebula, NGC 7023, is well suited for this study. We will use the measurements obtained from the proposed observations to study the processes that determine the ionization balance of C60 and hence other large aromatic molecules in space. This will significantly advance our understanding of the physics and chemistry of large aromatic molecules in the universe.

OBSERVING DESCRIPTION

We will use MIRI-MRS in order to cover the wavelength range required to identify the signatures of C60⁺ and C60⁺⁺ in the cavity of the reflection nebula NGC~7023. The regions where the corresponding molecular signatures peak are located at a distance of 5, 15, and 25 arcsec respectively from the star, and thus respective observations per instrument are required for each region. For each location we will use all 4 channels of MIRI MRS and their respective sub-bands. For the observations in all regions we employ the "Extended Source, 4-Point ALL" dither pattern. The FASTR1 and SLOWR1 readout patterns will be used for MIRI MRS IFU observations respectively.

We will perform dedicated background observations for each region matched to the readout pattern and number of groups per integration and integrations per exposure of the science pointing, and with a ``2-Point" dithering pattern optimized for background sources. We also request MIRI-MRS simultaneous imaging in order to improve astrometry following Jdix recommendations.

Proposal 8007 - Targets - Characterizing C60 cations in the cavity of NGC 7023

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	NGC7023-25	RA: 21 01 34.2695 (315.3927896d) Dec: +68 10 7.56 (68.16877d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Intracuster medium]</i> <i>Extended=YES</i>		
	(2)	NGC7023-15	RA: 21 01 35.3734 (315.3973892d) Dec: +68 10 0.66 (68.16685d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Intracuster medium]</i> <i>Extended=YES</i>		
	(3)	NGC7023-5	RA: 21 01 36.2500 (315.4010417d) Dec: +68 09 51.00 (68.16417d) Equinox: J2000 <i>Comments:</i> <i>Category=ISM</i> <i>Description=[Intracuster medium]</i> <i>Extended=YES</i>		
	(4)	NGC-7023-Bkg-25	RA: 21 01 42.9708 (315.4290450d) Dec: +68 04 9.30 (68.06925d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=ISM</i> <i>Description=[Interstellar molecules, Photodissociation regions, Reflection nebulae]</i> <i>Extended=YES</i>	Epoch of Position: 2000	
	(5)	NGC-7023-Bkg-5	RA: 21 01 42.9708 (315.4290450d) Dec: +68 04 9.30 (68.06925d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=ISM</i> <i>Description=[Interstellar molecules, Photodissociation regions, Reflection nebulae]</i> <i>Extended=YES</i>	Epoch of Position: 2000	
	(6)	NGC-7023-Bkg-15	RA: 21 01 42.9708 (315.4290450d) Dec: +68 04 9.30 (68.06925d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=ISM</i> <i>Description=[Interstellar molecules, Photodissociation regions, Reflection nebulae]</i> <i>Extended=YES</i>	Epoch of Position: 2000	

Proposal 8007 - Observation 1 - Characterizing C60 cations in the cavity of NGC 7023

Observation	Proposal 8007, Observation 1: MIRI-MRS 25												Mon Aug 11 23:00:09 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[MIRI-MRS-25-BKG (Obs 5)]												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates					Targ. Coord. Corrections			Miscellaneous		
	(1)	NGC7023-25	RA: 21 01 34.2695 (315.3927896d) Dec: +68 10 7.56 (68.16877d) Equinox: J2000										
	Comments: Category=ISM Description=[Intracluster medium] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				YES			BRIGHTSKY		Allow Auto Reorder		
Dithers	#	Dither Type					Optimized For				Direction		
	1	4-Point					EXTENDED SOURCE				NEGATIVE		
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F770W	FASTR1	5	15	1	Dither 1	4	60	308.04	
	1	SHORT(A)	MRSLONG		FASTR1	41	3	1	Dither 1	4	12	1387.52	223170
	1	SHORT(A)	MRSSHORT		FASTR1	41	3	1	Dither 1	4	12	1387.52	223170
	2		IMAGER	F1000W	FASTR1	5	15	1	Dither 1	4	60	308.04	
	2	MEDIUM(B)	MRSLONG		FASTR1	41	3	1	Dither 1	4	12	1387.52	223170
	2	MEDIUM(B)	MRSSHORT		FASTR1	41	3	1	Dither 1	4	12	1387.52	223170
	3		IMAGER	F1130W	FASTR1	5	15	1	Dither 1	4	60	308.04	
	3	LONG(C)	MRSLONG		FASTR1	41	3	1	Dither 1	4	12	1387.52	223170
	3	LONG(C)	MRSSHORT		FASTR1	41	3	1	Dither 1	4	12	1387.52	223170

Proposal 8007 - Observation 1 - Characterizing C60 cations in the cavity of NGC 7023

Special Requirements	Sequence Observations 1, 5, Non-interruptible
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Proposal 8007 - Observation 5 - Characterizing C60 cations in the cavity of NGC 7023

Observation	Proposal 8007, Observation 5: MIRI-MRS-25-BKG												Mon Aug 11 23:00:09 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [MIRI-MRS 25 (Obs 1)]													
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name		Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(4)	NGC-7023-Bkg-25		RA: 21 01 42.9708 (315.4290450d)				Epoch of Position: 2000						
				Dec: +68 04 9.30 (68.06925d)										
				Equinox: J2000										
		Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=ISM Description=[Interstellar molecules, Photodissociation regions, Reflection nebulae] Extended=YES												
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				YES			BRIGHTSKY			Allow Auto Reorder		
Dithers	#	Dither Type						Optimized For				Direction		
	1	2-Point						EXTENDED SOURCE				NEGATIVE		
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	FASTR1	5	10	1	Dither 1	2	20	102.103		
	1	SHORT(A)	MRSLONG		FASTR1	41	3	1	Dither 1	2	6	693.76		
	1	SHORT(A)	MRSSHORT		FASTR1	41	3	1	Dither 1	2	6	693.76		
	2		IMAGER	F1000W	FASTR1	5	10	1	Dither 1	2	20	102.103		
	2	MEDIUM(B)	MRSLONG		FASTR1	41	3	1	Dither 1	2	6	693.76		
	2	MEDIUM(B)	MRSSHORT		FASTR1	41	3	1	Dither 1	2	6	693.76		
	3		IMAGER	F1130W	FASTR1	5	10	1	Dither 1	2	20	102.103		
	3	LONG(C)	MRSLONG		FASTR1	41	3	1	Dither 1	2	6	693.76		
	3	LONG(C)	MRSSHORT		FASTR1	41	3	1	Dither 1	2	6	693.76		

Proposal 8007 - Observation 5 - Characterizing C60 cations in the cavity of NGC 7023

Special Requirements	Sequence Observations 1, 5, Non-interruptible
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Proposal 8007 - Observation 2 - Characterizing C60 cations in the cavity of NGC 7023

Observation	Proposal 8007, Observation 2: MIRI-MRS 15												Mon Aug 11 23:00:09 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[MIRI-MRS-15-BKG (Obs 6)]													
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(2)	NGC7023-15	RA: 21 01 35.3734 (315.3973892d) Dec: +68 10 0.66 (68.16685d) Equinox: J2000											
	Comments: Category=ISM Description=[Intracluster medium] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				YES			FULL			Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F1130W	FASTR1	5	46	1	Dither 1	4	184	3052.544		
	1	LONG(C)	MRSLONG		SLOWR1	32	1	1	Dither 1	4	4	3057.91	223170	
	1	LONG(C)	MRSSHORT		SLOWR1	32	1	1	Dither 1	4	4	3057.91	223170	
	2		IMAGER	F1000W	FASTR1	5	46	1	Dither 1	4	184	3052.544		
	2	MEDIUM(B)	MRSLONG		SLOWR1	32	1	1	Dither 1	4	4	3057.91	223170	
	2	MEDIUM(B)	MRSSHORT		SLOWR1	32	1	1	Dither 1	4	4	3057.91	223170	
	3		IMAGER	F770W	FASTR1	5	46	1	Dither 1	4	184	3052.544		
	3	SHORT(A)	MRSLONG		SLOWR1	32	1	1	Dither 1	4	4	3057.91	223170	
	3	SHORT(A)	MRSSHORT		SLOWR1	32	1	1	Dither 1	4	4	3057.91	223170	

Proposal 8007 - Observation 2 - Characterizing C60 cations in the cavity of NGC 7023

Special Requirements	Sequence Observations 2, 6, Non-interruptible
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Proposal 8007 - Observation 6 - Characterizing C60 cations in the cavity of NGC 7023

Observation	Proposal 8007, Observation 6: MIRI-MRS-15-BKG												Mon Aug 11 23:00:09 GMT 2025
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [MIRI-MRS 15 (Obs 2)]												
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(6)	NGC-7023-Bkg-15	RA: 21 01 42.9708 (315.4290450d) Dec: +68 04 9.30 (68.06925d) Equinox: J2000				Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=ISM												
	Description=[Interstellar molecules, Photodissociation regions, Reflection nebulae] Extended=YES												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
		All MRS				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	2-Point				EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		IMAGER	F1130W	FASTR1	5	46	1	Dither 1	2	92	1526.272	
	1	LONG(C)	MRSLONG		SLOWR1	32	1	1	Dither 1	2	2	1528.955	
	1	LONG(C)	MRSSHORT		SLOWR1	32	1	1	Dither 1	2	2	1528.955	
	2		IMAGER	F1000W	FASTR1	5	46	1	Dither 1	2	92	1526.272	
	2	MEDIUM(B)	MRSLONG		SLOWR1	32	1	1	Dither 1	2	2	1528.955	
	2	MEDIUM(B)	MRSSHORT		SLOWR1	32	1	1	Dither 1	2	2	1528.955	
	3		IMAGER	F770W	FASTR1	5	46	1	Dither 1	2	92	1526.272	
	3	SHORT(A)	MRSLONG		SLOWR1	32	1	1	Dither 1	2	2	1528.955	
	3	SHORT(A)	MRSSHORT		SLOWR1	32	1	1	Dither 1	2	2	1528.955	

Proposal 8007 - Observation 6 - Characterizing C60 cations in the cavity of NGC 7023

Special Requirements	Sequence Observations 2, 6, Non-interruptible
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Proposal 8007 - Observation 3 - Characterizing C60 cations in the cavity of NGC 7023

Observation	Proposal 8007, Observation 3: MIRI-MRS 5												Mon Aug 11 23:00:09 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[MIRI-MRS-5-BKG (Obs 7)]													
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name			Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(3)	NGC7023-5			RA: 21 01 36.2500 (315.4010417d) Dec: +68 09 51.00 (68.16417d) Equinox: J2000									
	Comments: Category=ISM Description=[Intracluster medium] Extended=YES													
Acquisition	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray			Grating Wheel Direction			
		All MRS			YES			BRIGHTSKY			Allow Auto Reorder			
Dithers	#	Dither Type					Optimized For				Direction			
	1	4-Point					EXTENDED SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	FASTR1	5	3	1	Dither 1	4	12	58.839		
	1	SHORT(A)	MRSLONG		FASTR1	38	4	1	Dither 1	4	16	1720.525	223170	
	1	SHORT(A)	MRSSHORT		FASTR1	38	4	1	Dither 1	4	16	1720.525	223170	
	2		IMAGER	F1000W	FASTR1	5	3	1	Dither 1	4	12	58.839		
	2	MEDIUM(B)	MRSLONG		FASTR1	38	4	1	Dither 1	4	16	1720.525	223170	
	2	MEDIUM(B)	MRSSHORT		FASTR1	38	4	1	Dither 1	4	16	1720.525	223170	
	3		IMAGER	F1130W	FASTR1	5	3	1	Dither 1	4	12	58.839		
	3	LONG(C)	MRSLONG		FASTR1	38	4	1	Dither 1	4	16	1720.525	223170	
	3	LONG(C)	MRSSHORT		FASTR1	38	4	1	Dither 1	4	16	1720.525	223170	

Proposal 8007 - Observation 3 - Characterizing C60 cations in the cavity of NGC 7023

Special Requirements	Sequence Observations 3, 7, Non-interruptible
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Proposal 8007 - Observation 7 - Characterizing C60 cations in the cavity of NGC 7023

Observation	Proposal 8007, Observation 7: MIRI-MRS-5-BKG												Mon Aug 11 23:00:09 GMT 2025	
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observation For: [MIRI-MRS 5 (Obs 3)]													
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(5)	NGC-7023-Bkg-5	RA: 21 01 42.9708 (315.4290450d) Dec: +68 04 9.30 (68.06925d) Equinox: J2000				Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.													
	Category=ISM													
	Description=[Interstellar molecules, Photodissociation regions, Reflection nebulae]													
Acquisition	Extended=YES													
	#	Target												
	1	NONE												
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray			Grating Wheel Direction		
		All MRS				YES			BRIGHTSKY			Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction				
	1	2-Point				EXTENDED SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1		IMAGER	F770W	FASTR1	5	1	1	Dither 1	2	2	8.653		
	1	SHORT(A)	MRSLONG		FASTR1	38	4	1	Dither 1	2	8	860.262		
	1	SHORT(A)	MRSSHORT		FASTR1	38	4	1	Dither 1	2	8	860.262		
	2		IMAGER	F1000W	FASTR1	5	1	1	Dither 1	2	2	8.653		
	2	MEDIUM(B)	MRSLONG		FASTR1	38	4	1	Dither 1	2	8	860.262		
	2	MEDIUM(B)	MRSSHORT		FASTR1	38	4	1	Dither 1	2	8	860.262		
	3		IMAGER	F1130W	FASTR1	5	1	1	Dither 1	2	2	8.653		
	3	LONG(C)	MRSLONG		FASTR1	38	4	1	Dither 1	2	8	860.262		
	3	LONG(C)	MRSSHORT		FASTR1	38	4	1	Dither 1	2	8	860.262		

Proposal 8007 - Observation 7 - Characterizing C60 cations in the cavity of NGC 7023

Special Requirements	Sequence Observations 3, 7, Non-interruptible
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